Exam Date & Time: 11-May-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Pattern Recognition [CSE-3274]

Marks: 50 **Duration: 180 mins.**

PART-A

Answer all the questions.

1) Compare between traditional learning vs. machine learning. Describe a standard pattern recognition system and support your explanation with a suitable diagram. (5)

A)

B) Define a 'Data Set' in the context of pattern recognition and list two essential characteristics that make a data set suitable for building reliable pattern recognition models. Furthermore, describe one potential issue that could arise when the data set does not possess these characteristics and how it could affect the model's performance. (3)

C) For the table below, Compute the proximity of nominal attributes- (2)

Roll No	Marks	Grades
1	96	A
2	87	B
3	83	B
4	96	A

2) A healthcare technology company has developed a machine learning model designed to predict patient adherence to prescribed medication regimens. The model outputs a probability that indicates how likely it is that a patient will follow their prescription plan. (5)

A) Adherence is critical as it directly impacts patient health outcomes and healthcare costs. Dataset is as follows-

Patient ID	Actual Adherence (A)	Predicted Probability (P)
1	Adhered	0.92
2	Not Adhered	0.81
3	Adhered	0.78
4	Adhered	0.56
5	Not Adhered	0.47
6	Adhered	0.65
7	Not Adhered	0.30
8	Adhered	0.85
9	Not Adhered	0.25
10	Adhered	0.70

Using the dataset provided, which contains the actual medication adherence status and the predicted probability of adherence for a set of 10 patients, plot an ROC curve to evaluate the effectiveness of the predictive model.

B) To study the performance of three detergents and three different water temperatures the following whiteness readings were obtained with specially designed equipment. Perform a two-way ANOVA, using 5% level of significance.

Water Temperature	A	B	C
Cold Water	47	45	50
Warm Water	39	42	52
Hot Water	44	36	48

(5)

3) (5)

A)

During a language learning study, participants are presented with words in two foreign languages, Language A and Language B. The success rate of remembering a word in Language A is denoted as $\hat{p}_A$ and in Language B as $\hat{p}_B$. Before the experiment, these success rates are hypothesized to be $\hat{p}_A = 0.65$ and $\hat{p}_B = 0.40$ based on the languages' perceived difficulty levels. Each participant goes through a series of recall tests where they are asked to remember and repeat the words. The results for each language, marked as successful recall ('Success', S) or unsuccessful recall ('Failure', F), are tracked over multiple trials.

Trial	Recall Pattern
1	S, F, S, F
2	F, S, F, S
3	S, S, F, F
4	F, F, S, S

For the above data, apply the EM Algorithm to update the probabilities $\hat{p}_A$ and $\hat{p}_B$ after one iteration.

B) Elaborate the fundamental concept of the Perceptron Learning Algorithm-
•Define the Perceptron and its operational mechanics as a binary classifier in machine learning.
•Describe the training process of the Perceptron, including the initialization of weights and the role of the learning rate.
•Illustrate the Perceptron Learning Rule and how it adjusts weights in response to classified training instances. (5)
•Explain how convergence is reached or the condition under which the Perceptron algorithm fails to converge.
•Discuss the limitations of a single-layer Perceptron and the kind of problems it can solve effectively.

4) Solve the below problem -

A)

Probabilities of each coin showing heads or tails

Coin considered	Probability of heads	Probability of tails
coin1	0.9	0.1
coin2	0.1	0.9

(5)

Obtain the probability for-
a) Generating sequesnce hhhthhhhth using coin1 and coin2.
b) Generating sequesnce tttttttttt using coin1 and coin2.
Also examine the principles and computational mechanics of a Hopfield network.

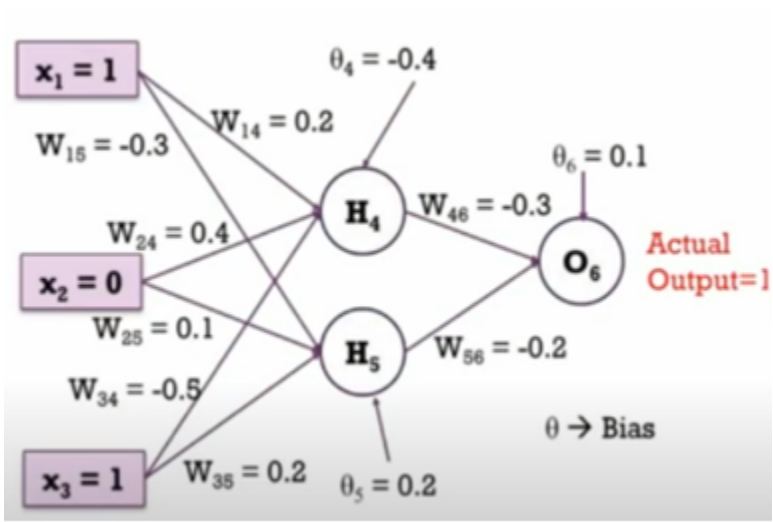
B) Apply K Medoid clustering algorithm to form 2 clusters on the dataset. (5)

Point	X	Y
1	1	4
2	5	1
3	5	2
4	5	4
5	10	4
6	25	4
7	25	6
8	25	7

Use Euclidean distance to find distance between data points and medoids.

5) Given that the neurons use a sigmoid activation function, conduct a forward pass and a backward pass on the neural network. Assume the target output for y is 1 and the learning rate is set at 0.9.

A)



(5)

B) Summarize the advancements that Deep Learning has brought to image classification tasks, citing specific neural network architectures that have been influential. Also discuss how Deep Learning has improved the accuracy and efficiency of Handwritten Digit Recognition application.
Evaluate the challenges faced when applying Deep Learning to pattern recognition and forecast the future of pattern recognition with Deep Learning, considering current research directions. (5)

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